

Bilkent University EEE202 LAB #2

Second-Order RLC Circuit's Transient Response to Step Input and Sinusoids

Preliminary Work

Consider the second-order RLC circuit shown in Figure 1. The component values are assigned as follows:

- Source resistance: $R = 50 \Omega$ (internal resistance of the signal generator)
- Inductor series resistance:

$$r_{\text{nom}} = \begin{cases} 0.1 \Omega, & L = 10 \mu\text{H} \\ 0.15 \Omega, & L = 15 \mu\text{H} \end{cases}$$

These values are specified at 10 kHz. The equivalent series resistance at the resonant frequency (f_r) should be calculated using

$$r = \sqrt{\frac{f_r}{10 \text{ kHz}}} r_{\text{nom}}.$$

- **The values of R_1 , C , L and f_r are assigned and should be checked from the Lab 2 RLC Assignment file.**

Note: Verify the assigned component values only if the specified frequency range is not satisfied or if a required component is unavailable. In such cases, contact `fuat.arslan@bilkent.edu.tr`.

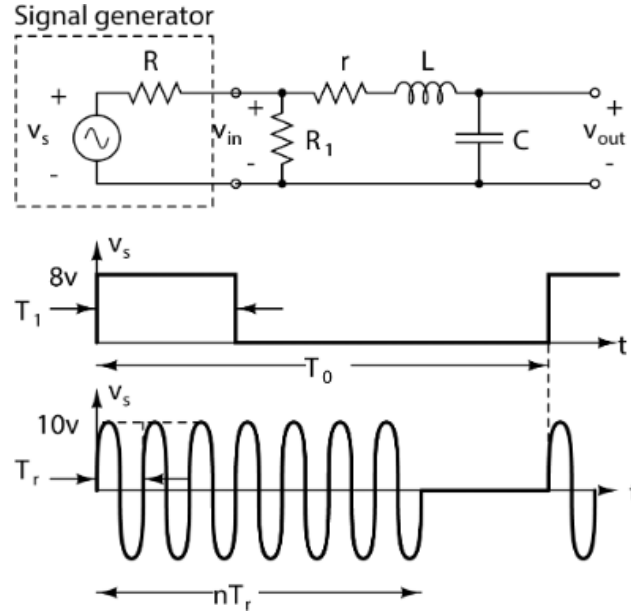


Figure 1: Second-order RLC circuit used in the experiment.

1. Calculate the quality factor Q of the circuit using the assigned component values. Note that Q depends on R , r , R_1 , L , and the resonant frequency f_r .
2. Calculate the response of the RLC circuit to a step input of magnitude 8 V applied at v_s . Derive an analytical expression valid for $t > 0$, plot the expected waveform using **MATLAB**, and verify the result using **LTSpice**.
3. Estimate the steady-state response of the RLC circuit to a sinusoidal excitation at frequency f_r with peak value 10 V, starting at $t = 0$:

$$v_s(t) = 10 u(t) \sin(2\pi f_r t)$$

Plot the expected waveform using **MATLAB**, and verify the result using **LTSpice**.

4. Using **LTSpice**, find the transient response of the circuit to the same sinusoidal excitation over the interval

$$0 \leq t \leq QT_r$$

assuming zero initial capacitor voltage and inductor current.

Experimental Work

Choose T_1 such that $T_1 > QT_r$ (at least $15T_r$), and choose $T_0 > 2T_1$. These conditions ensure that the circuit reaches steady state and that all initial conditions reset between repetitions.

Use a sinusoidal train of n sinusoids, which repeats itself every T_0 . Choose $n > Q$. nT_r is sufficiently large so that the output waveform reaches the steady state.

Measurement Setup

- Connect the **TRIG OUTPUT** of the signal generator to the **EXT TRIG** input of the oscilloscope using a coaxial cable.
- Set the oscilloscope trigger settings:
 - Source: **EXT**
 - Type: **Edge**
 - Slope: **Rising**
 - Mode: **Normal**
 - Coupling: **DC**
- Set the horizontal scale to approximately $T_1/10$ per division.

Step Input Excitation

To generate a step input of 8 V with width T_1 and repetition period T_0 , configure the signal generator (DS345) as follows:

1. Press [SHIFT] [DEFAULTS].
2. Press [FREQ] and set the frequency to $1/(2T_1)$.
3. Press [AMPL] and set the amplitude to $4V_{pp}$.
4. Press [OFFST] and set the offset to 2 V.
5. Press [PHASE] and set the phase to 180° .
6. Select the **Square Wave** function.
7. Enter **BURST** mode via the **SWEEP/MODULATE** menu.
8. Press [SHIFT] [TRIG SOURCE] and select **Internal RATE**.
9. Press [SHIFT] [TRIG RATE] and set the trigger rate to $1/T_0$.
10. Press [SHIFT] [BURST CNT] and set the burst count to 1.
11. Press [SWEEP ON/OFF] to enable burst mode.

Sinusoidal Excitation

To generate a burst of n sinusoidal cycles at frequency f_r :

1. Press [SHIFT] [DEFAULTS].
2. Press [FREQ] and set the frequency to f_r .
3. Press [AMPL] and set the amplitude to $10 V_{pp}$.
4. Enter **BURST** mode.
5. Press [SHIFT] [TRIG SOURCE] and select **Internal RATE**.
6. Press [SHIFT] [TRIG RATE] and set the trigger rate to $1/T_0$.
7. Press [SHIFT] [BURST CNT] and set the burst count to n , where $n > Q$.
8. Press [SWEEP ON/OFF] to enable burst mode.

Simultaneous Measurement of Input and Output

1. Connect the probe to the input of the RLC circuit.
2. Connect the probe to the output of the RLC circuit.
3. Keep horizontal scale and position unchanged to ensure proper superposition.

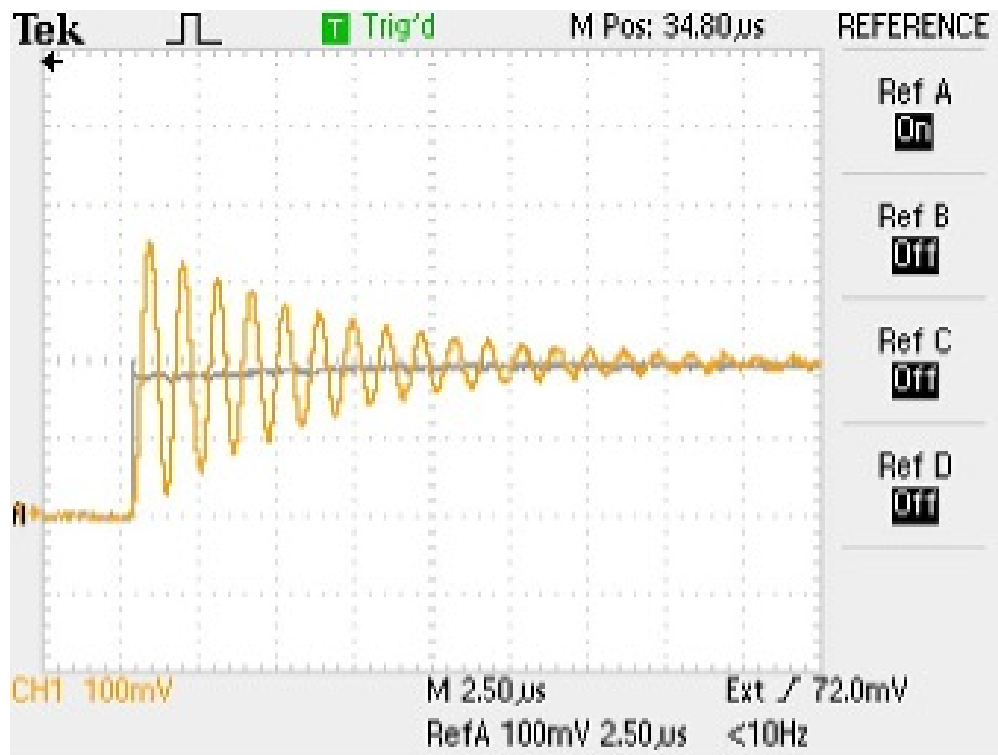


Figure 2: Measured input and output waveforms for step excitation.

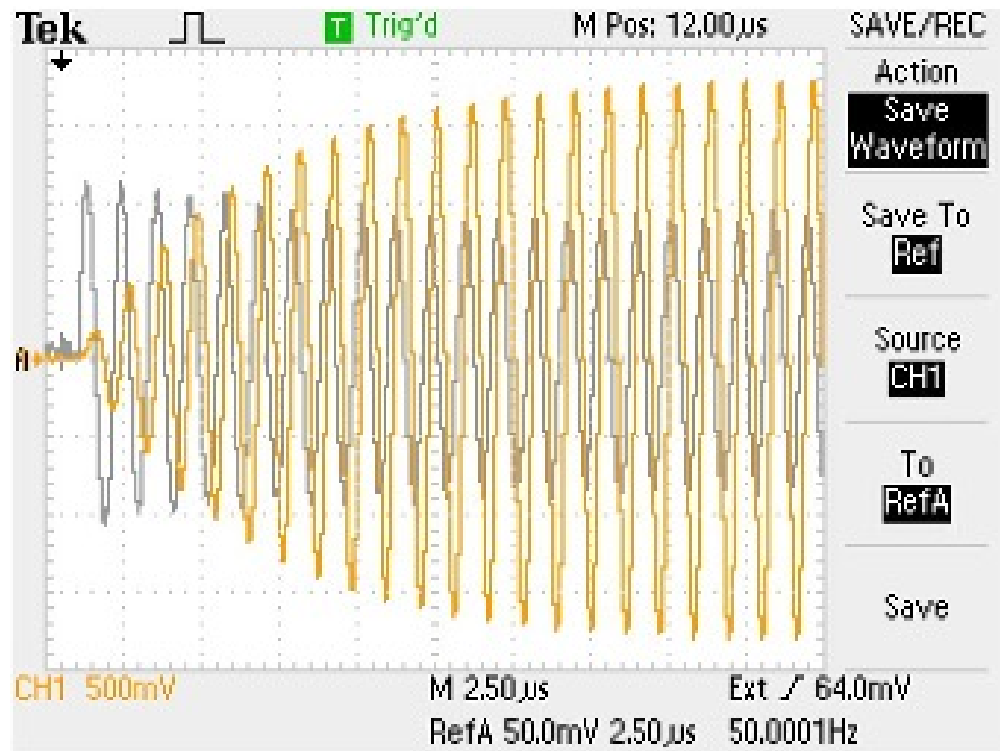


Figure 3: Measured input and output waveforms for sinusoidal excitation.

Compare and discuss the experimental results with the analytical and simulation results obtained in the preliminary work.

Note on experimental issues:

- Please prefer **polyester capacitors** over **ceramic capacitors**.
- The most common issues are related to **connections**, especially when using a **bread-board**. Therefore, **soldering is strongly advised**. Ensure that **all connections are secure**. At least you can tie them in midair. Sample implementation.